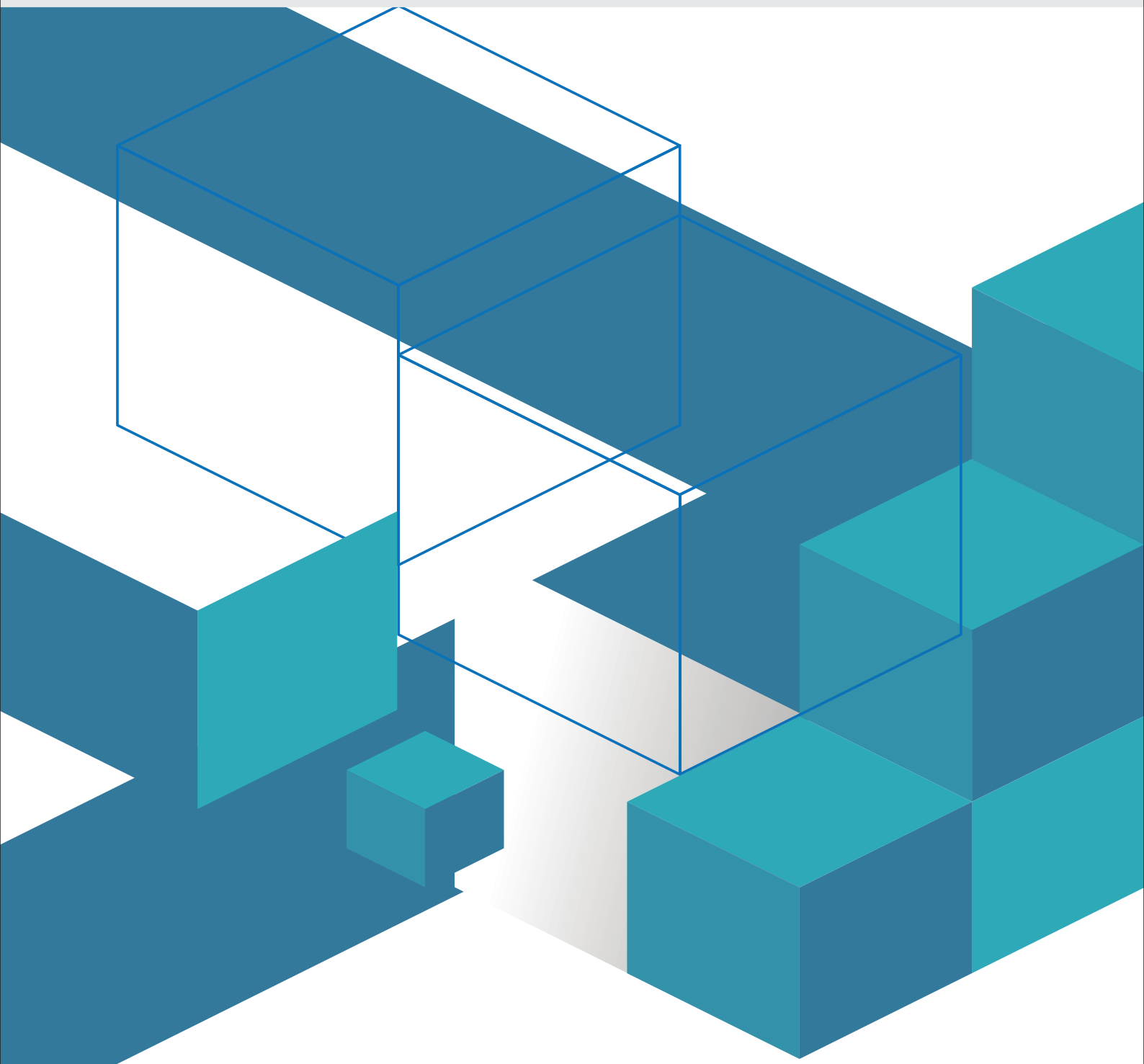
MATHEMATICS

DIFFERENTIAL EQUATION & AREA UNDER THE CURVE_MEGA_XII



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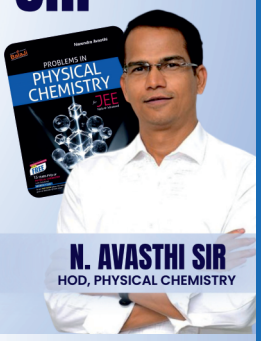

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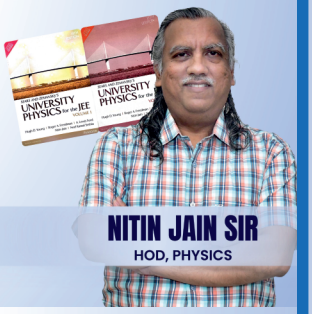


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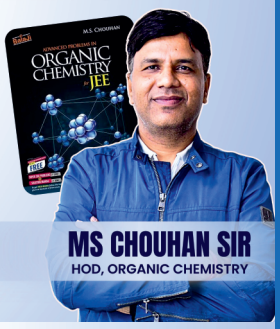


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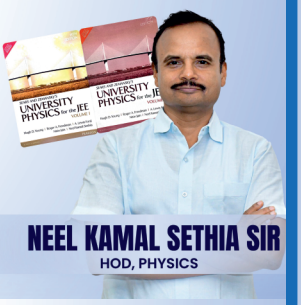


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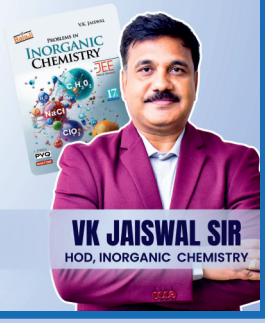


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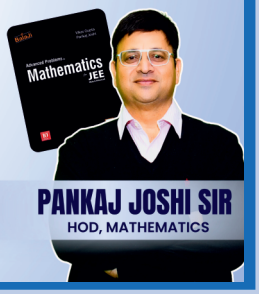


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MEGA-XII

DIFFERENTIAL EQUATION & AREA UNDER THE CURVE

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KEY CONCEPTS (DIFFERENTIAL EQUATION)
DIFFERENTIAL EQUATIONS OF FIRST ORDER AND FIRST DEGREE

DEFINITIONS :

1. An equation that involves independent and dependent variables and the derivatives of the dependent variables is called a **DIFFERENTIAL EQUATION**.

2. A differential equation is said to be ordinary, if the differential coefficients have reference to a single independent variable only and it is said to be **PARTIAL** if there are two or more independent variables. We are concerned with ordinary differential equations only. eg. $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=0$ is a partial differential equation.

3. Finding the unknown function is called **SOLVING OR INTEGRATING** the differential equation. The solution of the differential equation is also called its **PRIMITIVE**, because the differential equation can be regarded as a relation derived from it.

4. The order of a differential equation is the order of the highest differential coefficient occuring in it.

5. The degree of a differential equation which can be written as a polynomial in the derivatives is the degree of the derivative of the highest order occuring in it , after it has been expressed in a form free from radicals & fractions so far as derivatives are concerned, thus the differential equation :

$f(x , y)\left[\frac{d^m y}{dx^m}\right]^p + \phi(x , y)\left[\frac{d^{m-1}(y)}{dx^{m-1}}\right]^q + \ldots\ldots = 0$ is order m & degree p.

Note that in the differential equation $e^{y'''} - xy'' + y = 0$ order is three but degree doesn't apply.

6. FORMATION OF A DIFFERENTIAL EQUATION :

If an equation in independent and dependent variables having some arbitrary constant is given , then a differential equation is obtained as follows :

- ☞ Differentiate the given equation w.r.t. the independent variable (say x) as many times as the number of arbitrary constants in it .
- ☞ Eliminate the arbitrary constants .

The eliminant is the required differential equation . Consider forming a differential equation for $y^2 = 4a(x + b)$ where a and b are arbitary constant .

Note : A differential equation represents a family of curves all satisfying some common properties. This can be considered as the geometrical interpretation of the differential equation.

7. GENERAL AND PARTICULAR SOLUTIONS :

The solution of a differential equation which contains a number of independent arbitrary constants equal to the order of the differential equation is called the **GENERAL SOLUTION (OR COMPLETE INTEGRAL OR COMPLETE PRIMITIVE)** . A solution obtainable from the general solution by giving particular values to the constants is called a **PARTICULAR SOLUTION**.

Note that the general solution of a differential equation of the nth order contains 'n' & only 'n' independent arbitrary constants. The arbitrary constants in the solution of a differential equation are said to be independent, when it is impossible to deduce from the solution an equivalent relation containing fewer arbitrary constants. Thus the two arbitrary constants A, B in the equation $y = Ae^{x+B}$ are not independent since the equation can be written as $y = Ae^B . e^x = Ce^x$. Similarly the solution $y = A\sin x + B\cos(x + C)$ appears to contain three arbitrary constants, but they are really equivalent to two only.

8. Elementary Types Of First Order & First Degree Differential Equations .

TYPE-1. VARIABLES SEPARABLE : If the differential equation can be expressed as ;

$f(x)dx + g(y)dy = 0$ then this is said to be variable – separable type.

A general solution of this is given by $\int f(x)\,dx + \int g(y)\,dy = c$;

where c is the arbitrary constant . consider the example $(dy/dx) = e^{x-y} + x^2 . e^{-y}$.

Note : Sometimes transformation to the polar co–ordinates facilitates separation of variables. In this connection it is convenient to remember the following differentials.

If $x = r\cos\theta$; $y = r\sin\theta$ then,

(i) $x\,dx + y\,dy = r\,dr$ **(ii)** $dx^2 + dy^2 = dr^2 + r^2 d\theta^2$ **(iii)** $x\,dy - y\,dx = r^2 d\theta$

If $x = r\sec\theta$ & $y = r\tan\theta$ then $x\,dx - y\,dy = r\,dr$ and $x\,dy - y\,dx = r^2\sec\theta\,d\theta$.

TYPE-2 : $\frac{dy}{dx} = f(ax + by + c)$, $b \neq 0$.

To solve this, substitute $t = ax + by + c$. Then the equation reduces to separable type in the variable t and x which can be solved.

Consider the example $(x + y)^2 \frac{dy}{dx} = a^2$.

TYPE-3. HOMOGENEOUS EQUATIONS :

A differential equation of the form $\frac{dy}{dx} = \frac{f(x, y)}{\phi(x, y)}$

where $f(x , y)$ & $\phi(x , y)$ are homogeneous functions of x & y, and of the same degree, is called **HOMOGENEOUS**. This equation may also be reduced to the form $\frac{dy}{dx} = g\left(\frac{x}{y}\right)$ & is solved by putting $y = vx$ so that the dependent variable y is changed to another variable v, where v is some unknown function, the differential equation is transformed to an equation with variables separable. Consider

$\frac{dy}{dx} + \frac{y(x+y)}{x^2} = 0$.

TYPE-4. EQUATIONS REDUCIBLE TO THE HOMOGENEOUS FORM :

If $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$; where $a_1b_2 - a_2b_1 \neq 0$, i.e. $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$

then the substitution $x = u + h$, $y = v + k$ transform this equation to a homogeneous type in the new variables u and v where h and k are arbitrary constants to be chosen so as to make the given equation homogeneous which can be solved by the method as given in Type – 3. If

(i) $a_1b_2 - a_2b_1 = 0$, then a substitution $u = a_1x + b_1y$ transforms the differential equation to an equation with variables separable. and

(ii) $b_1 + a_2 = 0$, then a simple cross multiplication and substituting $d(xy)$ for $x\,dy + y\,dx$ & integrating term by term yields the result easily.

Consider $\frac{dy}{dx} = \frac{x - 2y + 5}{2x + y - 1}$; $\frac{dy}{dx} = \frac{2x + 3y - 1}{4x + 6y - 5}$ & $\frac{dy}{dx} = \frac{2x - y + 1}{6x - 5y + 4}$

(iii) In an equation of the form : $y f(xy)\,dx + x g(xy)dy = 0$ the variables can be separated by the substitution $xy = v$.

IMPORTANT NOTE :

(a) The function $f(x , y)$ is said to be a homogeneous function of degree n if for any real number $t (\neq 0)$, we have $f(tx , ty) = t^n f(x , y)$.

For e.g. $f(x , y) = ax^{2/3} + hx^{1/3} . y^{1/3} + by^{2/3}$ is a homogeneous function of degree 2/3.

(b) A differential equation of the form $\frac{dy}{dx} = f(x , y)$ is homogeneous if $f(x , y)$ is a homogeneous function of degree zero i.e. $f(tx , ty) = t^\circ f(x , y) = f(x , y)$. The function f does not depend on x & y separately but only on their ratio $\frac{y}{x}$ or $\frac{x}{y}$.

LINEAR DIFERENTIAL EQUATIONS :

A differential equation is said to be linear if the dependent variable & its differential coefficients occur in the first degree only and are not multiplied together.

The nth order linear differential equation is of the form ;

$a_0(x)\frac{d^ny}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + ... + a_n(x) . y = \phi(x)$. Where $a_0(x)$, $a_1(x)$ ..... $a_n(x)$ are called the coefficients of the differential equation.

Note that a linear differential equation is always of the first degree but every differental equation of the first degree need not be linear. e.g. the differential equation $\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3 + y^2 = 0$ is not linear, though its degree is 1.

TYPE-5. LINEAR DIFFERENTIAL EQUATIONS OF FIRST ORDER :

The most general form of a linear differential equations of first order is $\frac{dy}{dx} + Py = Q$, where P & Q are functions of x .

To solve such an equation multiply both sides by $e^{\int Pdx}$.

NOTE :

(1) The factor $e^{\int Pdx}$ on multiplying by which the left hand side of the differential equation becomes the differential coefficient of some function of x & y , is called integrating factor of the differential equation popularly abbreviated as I. F.

(2) It is very important to remember that on multiplying by the integrating factor , the left hand side becomes the derivative of the product of y and the I. F.

(3) Some times a given differential equation becomes linear if we take y as the independent variable and x as the dependent variable. e.g. the equation ;

$(x + y + 1)\frac{dy}{dx} = y^2 + 3$ can be written as $(y^2 + 3)\frac{dx}{dy} = x + y + 1$ which is a linear differential equation.

TYPE- 6. EQUATIONS REDUCIBLE TO LINEAR FORM :

The equation $\frac{dy}{dx} + py = Q \cdot y^n$ where P & Q functions of x , is reducible to the linear form by dividing it by y^n & then substituting $y^{-n+1} = Z$. Its solution can be obtained as in **Type-5**. Consider the example $(x^3y^2 + xy)\,dx = dy$.

The equation $\frac{dy}{dx} + Py = Q \cdot y^n$ is called **BERNOULI'S EQUATION**.

9. TRAJECTORIES :

Suppose we are given the family of plane curves.

$\Phi(x, y, a) = 0$

depending on a single parameter a.

A curve making at each of its points a fixed angle α with the curve of the family passing through that point is called an ***isogonal trajectory*** of that family ; if in particular $\alpha = \pi/2$, then it is called an ***orthogonal trajectory***.

Orthogonal trajectories : We set up the differential equation of the given family of curves. Let it be of the form

$F(x, y, y') = 0$

The differential equation of the orthogonal trajectories is of the form

$$F\left(x, y, -\frac{1}{y'}\right) = 0$$

The general integral of this equation

$\Phi_1(x, y, C) = 0$

gives the family of orthogonal trajectories.

Note : Following exact differentials must be remembered :

(i) $xdy + y\,dx = d(xy)$ **(ii)** $\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$

(iii) $\frac{ydx - xdy}{y^2} = d\left(\frac{x}{y}\right)$ **(iv)** $\frac{xdy + ydx}{xy} = d(\ln xy)$

(v) $\frac{dx + dy}{x + y} = d(\ln(x + y))$ **(vi)** $\frac{xdy - ydx}{xy} = d\left(\ln\frac{y}{x}\right)$

(vii) $\frac{ydx - xdy}{xy} = d\left(\ln\frac{x}{y}\right)$ **(viii)** $\frac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$

(ix) $\frac{ydx - xdy}{x^2 + y^2} = d\left(\tan^{-1}\frac{x}{y}\right)$ **(x)** $\frac{xdx + ydy}{x^2 + y^2} = d\left[\ln\sqrt{x^2 + y^2}\right]$

(xi) $d\left(-\frac{1}{xy}\right) = \frac{xdy + ydx}{x^2y^2}$ **(xii)** $d\left(\frac{e^x}{y}\right) = \frac{ye^x dx - e^x dy}{y^2}$

(xiii) $d\left(\frac{e^y}{x}\right) = \frac{xe^y dy - e^y dx}{x^2}$

EXERCISE–I

(FORMATION & VARIABLES SEPARABLE)

1. State the order and degree of the following differential equations:

(i) $\left[\frac{d^2x}{dt^2}\right]^3 + \left[\frac{dx}{dt}\right]^4 - xt = 0$ (ii) $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$

2. Obtain the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$; where g , f & c are arbitary constants.

3. Form the differential equation of the family of curves represented by, $c(y + c)^2 = x^3$; where c is any arbitrary constant.

4. $\frac{\ell n(\sec x + \tan x)}{\cos x}\,dx = \frac{\ell n(\sec y + \tan y)}{\cos y}\,dy$

5. $\frac{dy}{dx} + \frac{\sqrt{(x^2 - 1)(y^2 - 1)}}{xy} = 0$ **6.** $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$

7. $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$ **8.** $\frac{dy}{dx} = \frac{x(2\ln x + 1)}{\sin y + y\cos y}$

9. It is known that the decay rate of radium is directly proportional to its quantity at each given instant. Find the law of variation of a mass of radium as a function of time if at $t = 0$, the mass of the radius was m_0 and during time t_0 α % of the original mass of radium decay.

10. The population P of a town decreases at a rate proportional to the number by which the population exceeds 1000, proportionality constant being k > 0. Find
(a) Population at any time t, given initial population of the town being 2500.
(b) If 10 years later the population has fallen to 1900, find the time when the population will be 1500.
(c) Predict about the population of the town in the long run.

11. $e^{(dy/dx)} = x + 1$ given that when $x = 0$, $y = 3$

12. A normal is drawn at a point P(x , y) of a curve. It meets the x– axis at Q. If PQ is of constant length k, then show that the differential equation describing such curves is, $y\dfrac{dy}{dx} = \pm\sqrt{k^2 - y^2}$. Find the equation of such a curve passing through (0, k).

13. Obtain the differential equation associated with the primitive ,
$y = c_1e^{3x} + c_2e^{2x} + c_3e^{x}$, where c_1, c_2, c_3 are arbitrary constants.

14. Find the curve y = f (x) where $f(x) \geq 0$, f (0) = 0 , bounding a curvilinear trapezoid with the base [0, x] whose area is proportional to $(n + 1)^{th}$ power of f (x). It is known that f (1) = 1.

15. $\dfrac{x\,dx - y\,dy}{x\,dy - y\,dx} = \sqrt{\dfrac{1 + x^2 - y^2}{x^2 - y^2}}$

16. A curve is such that the length of the polar radius of any point on the curve is equal to the length of the tangent drawn at this point. Form the differential equation and solve it to find the equation of the curve.

EXERCISE–II

(HOMOGENEOUS)

1. (a) $\dfrac{dy}{dx} = \dfrac{x^2 + xy}{x^2 + y^2}$ (b) $(x^3 - 3xy^2)\,dx = (y^3 - 3x^2y)\,dy$

2. Find the equation of a curve such that the projection of its ordinate upon the normal is equal to its abscissa.

3. The light rays emanating from a point source situated at origin when reflected from the mirror of a search light are reflected as beam parallel to the x-axis. Show that the surface is parabolic, by first forming the differential equation and then solving it.

4. The perpendicular from the origin to the tangent at any point on a curve is equal to the abscissa of the point of contact. Find the equation of the curve satisfying the above condition and which passes through (1, 1).

5. Show that the equation of the curve intersecting with the x- axis at the point x = 1 and for which the length of the subnormal at any point of the curve is equal to the arthemetic mean of the co-ordinates of this point is $(y - x)^2(x + 2y) = 1$.

6. Use the substitution $y^2 = a - x$ to reduce the equation $y^3 \cdot \dfrac{dy}{dx} + x + y^2 = 0$ to homogeneous form and hence solve it.

7. $\left[x\cos\dfrac{y}{x} + y\sin\dfrac{y}{x}\right]y = \left[y\sin\dfrac{y}{x} - x\cos\dfrac{y}{x}\right]x\dfrac{dy}{dx}$

8. Find the curve for which any tangent intersects the y–axis at the point equidistant from the point of tangency and the origin.

9. $(x - y)\,dy = (x + y + 1)\,dx$ **10.** $\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$ **11.** $\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}$

12. $\frac{dy}{dx} = \frac{x + y + 1}{2x + 2y + 3}$ **13.** $\frac{dy}{dx} = \frac{2(y + 2)^2}{(x + y - 1)^2}$

14. $\frac{dy}{dx} + \frac{\cos x\,(3\cos y - 7\sin x - 3)}{\sin y\,(3\sin x - 7\cos y + 7)} = 0$

15. Show that the curve such that the distance between the origin and the tangent at an arbitrary point is equal to the distance between the origin and the normal at the same point,

$$\sqrt{x^2 + y^2} = c\,e^{\pm \tan^{-1}\frac{y}{x}}$$

EXERCISE–III

(LINEAR)

1. Show that the equation of the curve whose slope at any point is equal to $y + 2x$ and which pass through the origin is $y = 2(e^x - x - 1)$.

2. Find the curve such that the area of the trapezium formed by the co–ordinate axes, ordinate of an arbitrary point & the tangent at this point equals half the square of its abscissa .

3. $x(x - 1)\frac{dy}{dx} - (x - 2)y = x^3(2x - 1)$ **4.** $(1 + y + x^2y)\,dx + (x + x^3)\,dy = 0$

5. Find the curve possessing the property that the intercept , the tangent at any point of a curve cuts off on the y–axis is equal to the square of the abscissa of the point of tangency.

6. $\sin x\,\frac{dy}{dx} + 3y = \cos x$ **7.** $x\,\frac{dy}{dx} - y = 2x^2 \operatorname{cosec} 2x$

8. $(1 + y^2)\,dx = (\tan^{-1} y - x)dy$

9. Find the curve such that the area of the rectangle constructed on the abscissa of any point and the initial ordinate of the tangent at this point is equal to a^2. (Initial ordinate means y intercept of the tangent).

10. Let the function $\ell n\, f(x)$ is defined where $f(x)$ exists for $x \geq 2$ & k is fixed positive real number, prove that if $\frac{d}{dx}(x \cdot f(x)) \leq -k\,f(x)$ then $f(x) \leq A\,x^{-1-k}$ where A is independent of x.

11. Find the differentiable function which satisfies the equation $f(x) = -\int_0^x f(t)\tan t\,dt + \int_0^x \tan(t - x)\,dt$

where $x \in (-\pi/2, \pi/2)$

12. Find all functions $f(x)$ defined on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with real values and has a primitive $F(x)$ such that

$f(x) + \cos x \cdot F(x) = \frac{\sin 2x}{(1 + \sin x)^2}$. Find $f(x)$.

13. If y_1 & y_2 be solutions of the differential equation $\frac{dy}{dx} + Py = Q$, where P & Q are functions of x alone, and $y_2 = y_1 z$, then prove that

$z = 1 + a e^{-\int \frac{Q}{y_1} dx}$, 'a' being an arbitrary constant.

14. $\frac{dy}{dx} + \frac{y}{x} \ln y = \frac{y}{x^2} (\ln y)^2$

15. $\frac{dy}{dx} + xy = y^2 e^{x^2/2} . \sin x$

16. $2\frac{dy}{dx} - y \sec x = y^3 \tan x$

17. $x^2 y - x^3 \frac{dy}{dx} = y^4 \cos x$

18. $y(2xy + e^x) dx - e^x dy = 0$

19. Find the curve for which the area of the triangle formed by the x–axis, the tangent line and radius vector of the point of tangency is equal to a^2.

20. A tank contains 100 litres of fresh water. A solution containing 1 gm/litre of soluble lawn fertilizer runs into the tank at the rate of 1 lit/min, and the mixture is pumped out of the tank at the rate of 3 litres/min. Find the time when the amount of fertilizer in the tank is maximum.

21. $(x^3 + y^2 + 2) dx + 2y dy = 0$

22. $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1 + x) e^x \sec y$

23. $\left(\frac{dy}{dx}\right)^2 - (x+y)\frac{dy}{dx} + xy = 0$

24. $(1 - xy + x^2 y^2) dx = x^2 dy$

25. $y y' \sin x = \cos x (\sin x - y^2)$

EXERCISE–IV

1. The order and degree of the differential equation $\left(1 + 3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are **[AIEEE 2002]**

(A) $\left(1, \frac{2}{3}\right)$ (B) (3, 1) (C) (3, 3) (D) (1, 2)

2. The solution to the equation $\frac{d^2y}{dx^2} = e^{-2x}$ is : **[AIEEE 2002]**

(A) $\frac{2^{-2x}}{4}$ (B) $\frac{e^{-2x}}{4} + cx + d$ (C) $\frac{1}{4} e^{-2x} + cx^2 + d$ (D) $\frac{1}{4} e^{-4x} + cx + d$

3. The degree and order of the differential equation of the family of all parabolas whose axis is x - axis, are respectively **[AIEEE 2003]**

(A) 2 , 3 (B) 2, 1 (C) 1, 2 (D) 3, 2

4. The solution of the differential equation $(1 + y^2) + (x - e^{\tan^{-1} y})\frac{dy}{dx} = 0$, is **[AIEEE 2003]**

(A) $x e^{2\tan^{-1} y} = e^{\tan^{-1} y} + k$ (B) $(x - 2) = k e^{-\tan^{-1} y}$

(C) $2x e^{\tan^{-1} y} = e^{2\tan^{-1} y} + k$ (D) $x e^{\tan^{-1} y} = \tan^{-1} y + k$

5. The differential equation for the family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is **[AIEEE 2004]**

(A) $2(x^2 - y^2)\, y' = xy$ (B) $2(x^2 + y^2)\, y' = xy$

(C) $(x^2 - y^2)\, y' = 2xy$ (D) $(x^2 + y^2)\, y' = 2xy$

6. The solution of the differential equation, $ydx + (x + x^2 y)\, dy = 0$ is **[AIEEE 2004]**

(A) $\frac{1}{xy} = C$ (B) $-\frac{1}{xy} + \log y = C$ (C) $\frac{1}{xy} + \log y = C$ (D) $\log y = Cx$

7. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where $c > 0$ is a parameter, is of order and degree as follows **[AIEEE- 2005]**

(A) order 1, degree 2 (B) order 1, degree 1

(C) order 1, degree 3 (D) order 2, degree 2

8. If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is **[AIEEE-2005]**

(A) $y \log\left(\frac{x}{y}\right) = cx$ (B) $x \log\left(\frac{y}{x}\right) = cy$ (C) $\log\left(\frac{y}{x}\right) = cx$ (D) $\log\left(\frac{x}{y}\right) = cy$

9. The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants, is of **[AIEEE 2006]**

(A) first order and second degree (B) first order and first degree

(C) second order and first degree (D) second order and second degree

10. The differential equation of all circles passing through the origin and having their centres on the x-axis is

(A) $x^2 = y^2 + xy\frac{dy}{dx}$ (B) $x^2 = y^2 + 3xy\frac{dy}{dx}$ **[AIEEE 2007]**

(C) $y^2 = x^2 + 2xy\frac{dy}{dx}$ (D) $y^2 = x^2 - 2xy\frac{dy}{dx}$

11. The differential equation of the family of circles with fixed radius 5 units and centre on the line $y = 2$ is

(A) $(y - 2)\, y'^2 = 25 - (y - 2)^2$ (B) $(y - 2)^2\, y'^2 = 25 - (y - 2)^2$ **[AIEEE 2008]**

(C) $(x - 2)^2\, y'^2 = 25 - (y - 2)^2$ (D) $(x - 2)\, y'^2 = 25 - (y - 2)^2$

12. The solution of the differential equation $\frac{dy}{dx} = \frac{x + y}{x}$ satisfying the condition $y(1) = 1$ is **[AIEEE 2008]**

(A) $y = x\,\ell n\, x + x^2$ (B) $y = xe^{(x-1)}$ (C) $y = x\,\ell n\, x + x$ (D) $y = \ell n\, x + x$

13. The differential equation which represents the family of curves $y = c_1 e^{c_2 x}$ where c_1 and c_2 are arbitrary constants, is **[AIEEE 2009]**

(A) $y' = y^2$ (B) $y'' = y'\, y$ (C) $yy'' = y'$ (D) $yy'' = (y')^2$

14. The solution to the differential equation $\cos x\, dy = y(\sin x - y)\, dx$, $0 < x < \frac{\pi}{2}$, is **[AIEEE 2010]**

(A) $\tan x = (\sec x + c)y$ (B) $\sec x = (\tan x + c)y$

(C) $y\sec x = \tan x + c$ (D) $y\tan x = \sec x + c$

15. Let I be the purchase value of an equipment and V(t) be the value after it has been used for t years. The value V(t) depreciates at a rate given by differential equation $\frac{dV(t)}{dt} = -k(T-t)$, where $k > 0$ is a constant and T is the total life in years of the equipment. Then the scrap value V(T) of the equipment is : **[AIEEE 2011]**

(A) e^{-kT} (B) $T^2 - \frac{I}{k}$ (C) $I - \frac{kT^2}{2}$ (D) $I - \frac{k(T-t)^2}{2}$

16. If $\frac{dy}{dx} = y + 3 > 0$ and $y(0) = 2$, then $y(\ln 2)$ is equal to **[AIEEE 2011]**

(A) –2 (B) 7 (C) 5 (D) 13

17. The population p(t) at time t of a certain mouse species satisfies the differential equation **[AIEEE 2012]** $\frac{dp(t)}{dt} = 0.5\,p(t) - 450$. If $p(0) = 850$, then the time at which the population becomes zero is :

(A) 2 ln 18 (B) ln9 (C) $\frac{1}{2}$ ln 18 (D) ln 18

18. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is : **[JEE Main 2013]**

(A) 4500 (B) 2500 (C) 3000 (D) 3500

19. Let the population of rabbits surviving at a time t be governed by the differential equation $\frac{dp(t)}{dt} = \frac{1}{2}p(t) - 200$. If $p(0) = 100$, then p(t) equals **[JEE Main 2014]**

(A) $400 - 300\,e^{-t/2}$ (B) $400 - 300\,e^{t/2}$ (C) $300 - 200\,e^{-t/2}$ (D) $600 - 500\,e^{t/2}$

20. Let y(x) be the solution of the differential equation $(x \log x)\frac{dy}{dx} + y = 2x \log x$, $(x \geq 1)$. Then y(e) is equal to: **[JEE Main 2015]**

(A) 2e (B) e (C) 0 (D) 2

21. If a curve y = f(x) passes through the point (1, –1) and satisfies the differential equation, $y(1 + xy)\,dx = x\,dy$, then $f\left(-\frac{1}{2}\right)$ is equal to : **[JEE Main 2016]**

(A) $-\frac{2}{5}$ (B) $-\frac{4}{5}$ (C) $\frac{2}{5}$ (D) $\frac{4}{5}$

22. If $(2 + \sin x)\frac{dy}{dx} + (y + 1)\cos x = 0$ and $y(0) = 1$ then $y\left(\frac{\pi}{2}\right)$ is equal to : **[JEE Main 2017]**

(A) $\frac{4}{3}$ (B) $\frac{1}{3}$ (C) $-\frac{2}{3}$ (D) $-\frac{1}{3}$

23. Let y = y(x) be the solution of the differential equation

$\sin x\frac{dy}{dx} + y\cos x = 4x,\ x \in (0, \pi)$. If $y\left(\frac{\pi}{2}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to **[JEE Main 2018]**

(A) $-\frac{4}{9}\pi^2$ (B) $\frac{4}{9\sqrt{3}}\pi^2$ (C) $\frac{-8}{9\sqrt{3}}\pi^2$ (D) $-\frac{8}{9}\pi^2$

24. If y = y(x) is the solution of the differential equation, $x\frac{dy}{dx} + 2y = x^2$ satisfying y(1) = 1, then $y\left(\frac{1}{2}\right)$ is equal to **[IIT Main Online-2019]**

(A) $\frac{49}{16}$ (B) $\frac{13}{16}$ (C) $\frac{7}{64}$ (D) $\frac{1}{4}$

25. Let f : [0, 1] $\to$ R be such that f(x y) = f(x).f(y), for all x, y $\in$ [0, 1], and f(0) $\neq$ 0. If y = y(x) satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with y(0) = 1, then $y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is equal to :

[IIT Main Online-2019]

(A) 3 (B) 5 (C) 2 (D) 4

26. If $\frac{dy}{dx} + \frac{3}{\cos^2 x}y = \frac{1}{\cos^2 x}, x \in \left(\frac{-\pi}{3}, \frac{\pi}{3}\right)$, and $y\left(\frac{\pi}{4}\right) = \frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals :

[IIT Main Online-2019]

(A) $-\frac{4}{3}$ (B) $\frac{1}{3} + e^3$ (C) $\frac{1}{3}$ (D) $\frac{1}{3} + e^6$

27. Let f be a differentiable function such that $f'(x) = 7 - \frac{3}{4}\frac{f(x)}{x}, (x > 0)$ and $f(1) \neq 4$. Then $\lim_{x\to 0^+} xf\left(\frac{1}{x}\right)$

[IIT Main Online-2019]

(A) Exists and equals 0 (B) Exists and equals $\frac{4}{7}$

(C) Exists and equals 4 (D) Does not exist

28. The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2)dx + 2xy\ dy = 0$ which passes through (1, 1), is : **[IIT Main Online-2019]**

(A) A circle with centre on the x-axis

(B) A circle with centre on the y-axis

(C) A hyperbola with transverse axis along the x-axis

(D) An ellipse with major axis along the y-axis

29. If y(x) is the solution of the differential equation $\frac{dy}{dx}+\left(\frac{2x+1}{x}\right)y = e^{-2x}, x > 0,$ where $y(1) = \frac{1}{2}e^{-2}$, then :

[IIT Main Online-2019]

(A) $y(\log_e 2) = \frac{\log_e 2}{4}$

(B) y(x) is decreasing in (0, 1)

(C) y(x) is decreasing in $\left(\frac{1}{2},1\right)$

(D) $y(\log_e 2) = \log_e 4$

30. The solution of the differential equation, $\frac{dy}{dx} = (x-y)^2$, when y(1) = 1, is :**[IIT Main Online-2019]**

(A) $\log_e\left|\frac{2-y}{2-x}\right| = 2(y-1)$

(B) $-\log_e\left|\frac{1+x-y}{1-x+y}\right| = x+y-2$

(C) $\log_e\left|\frac{2-x}{2-y}\right| = x-y$

(D) $-\log_e\left|\frac{1-x+y}{1+x-y}\right| = 2(x-1)$

31. Let y = y(x) be the solution of the differential equation, $x\frac{dy}{dx} + y = x\log_e x$, (x > 1). If $2y(2) = \log_e 4 - 1$, then y(e) is equal to : **[IIT Main Online-2019]**

(A) $-\frac{e^2}{2}$ (B) $\frac{e^2}{4}$ (C) $-\frac{e}{2}$ (D) $\frac{e}{4}$

32. If a curve passes through the point (1, –2) and has slope of the tangent at any point (x, y) on it as $\frac{x^2-2y}{x}$, then the curve also passes through the point : **[IIT Main Online-2019]**

(A) $(\sqrt{3},0)$ (B) (3, 0) (C) $\left(-\sqrt{2},1\right)$ (D) (–1, 2)

33. If y = y(x) is the solution of the differential equation, $e^y\left(\frac{dy}{dx}-1\right) = e^x$ such that y(0) = 0, then y(1) is equal to : **[IIT Main Online-2020]**

(A) $2 + \log_e 2$ (B) 2e (C) $\log_e 2$ (D) $1 + \log_e 2$

34. Let y = y(x) be the solution curve of the differential equation, $(y^2-x)\frac{dy}{dx} = 1$, satisfying y(0) = 1. This curve intersects the x-axis at a point whose abscissa is : **[IIT Main Online-2020]**

(A) –e (B) 2 + e (C) 2 (D) 2 –e

35. Let y = y (x) be a solution of the differential equation, $\sqrt{1-x^2}\frac{dy}{dx} + \sqrt{1-y^2} = 0, |x| < 1.$

If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to : **[IIT Main Online-2020]**

(A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$ (C) $-\frac{1}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{2}}$

36. The differential equation of the family of curves, $x^2 = 4b(y+b)$, $b \varepsilon R$, is: **[IIT Main Online-2020]**

(A) $x(y')^2 = x - 2yy'$ (B) $x(y')^2 = 2yy' - x$ (C) $x(y')^2 = x + 2yy'$ (D) $xy'' = y'$

37. If $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$; $y(1) = 1$, then a value of x satisfying $y(x) = e$ is **[IIT Main Online-2020]**

(A) $\sqrt{3}e$ (B) $\frac{e}{\sqrt{2}}$ (C) $\sqrt{2}e$ (D) $\frac{1}{2}\sqrt{3}e$

38. If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation, $(x + 1)dy = ((x + 1)^2 + y - 3)\,dx$, $y(2) = 0$, then $y(3)$ is equal to _____. **[IIT Main Online-2020]**

EXERCISE–V

(PROBLEMS ASKED IN JEE & REE)

1. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true? **[JEE Advanced 2015]**

(A) $P = y + x$ (B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$ (D) $P - Q = x + y - y' - (y')^2$

2. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4)\frac{dy}{dx} - y^2 = 0$, $x > 0$, passes through the point (1, 3). Then the solution curve **[JEE Advanced 2016]**

(A) intersects $y = x + 2$ exactly at one point (B) intersects $y = x + 2$ exactly at two points

(C) intersects $y = (x + 2)^2$ (D) does **NOT** intersect $y = (x + 3)^2$

3. If $y = y(x)$ satisfies the differential equation **[JEE Advanced 2017]**

$$8\sqrt{x}\left(\sqrt{9+\sqrt{x}}\right)dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx, \quad x > 0$$

and $y(0) = \sqrt{7}$, then $y(256) =$

(A) 80 (B) 3 (C) 16 (D) 9

4. Let $f : \mathbf{R} \to \mathbf{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation

$\frac{dy}{dx} = (2 + 5y)(5y - 2)$, then the value of $\lim_{x \to -\infty} f(x)$ is _____. **[JEE Advanced 2018]**

5. Let Γ denote a curve $y = y(x)$ which is in the first quadrant and let the point (1, 0) lie on it. Let the tangent to Γ at a point P intersect the y-axis at Y_P. If PY_P has length 1 for each point P on Γ, then which of the following options is/are correct? **[JEE Advanced 2019]**

(A) $xy' - \sqrt{1 - x^2} = 0$ (B) $y = \log_e\left(\frac{1 + \sqrt{1 - x^2}}{x}\right) - \sqrt{1 - x^2}$

(C) $xy' + \sqrt{1 - x^2} = 0$ (D) $y = -\log_e\left(\frac{1 + \sqrt{1 - x^2}}{x}\right) + \sqrt{1 - x^2}$

6. Let *b* be a nonzero real number. Suppose f : **R** $\to$ **R** is a differentiable function such that f(0) = 1. If the derivative f ′ of f satisfies the equation

$$f'(x) = \frac{f(x)}{b^2 + x^2}$$

for all $x \in$ **R**, then which of the following statements is/are TRUE? **[JEE Advanced 2020]**

(A) If b > 0, then f is an increasing function

(B) If b < 0, then f is a decreasing function

(C) f(x) f(–x) = 1 for all $x \in R$

(D) f(x) – f(–x) = 0 for all $x \in R$

7. For any real numbers α and β, let $y_{\alpha,\beta}(x)$, $x \in R$, be the solution of the differential equation

$$\frac{dy}{dx} + ay = xe^{\beta x}, y(1) = 1$$

Let S = $\{y_{\alpha,\beta}(x) : \alpha, \beta \in R\}$. Then which of the following functions belong(s) to the set S ?

[JEE Advanced 2021]

(A) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(B) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(C) $f(x) = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(D) $f(x) = \frac{e^x}{2}\left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$

8. For $x \in$ **R**, let the function y(x) be the solution of the differential equation

$$\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right), \quad y(0) = 0.$$

Then, which of the following statements is/are **TRUE**? **[JEE Advanced 2022]**

(A) y(x) is an increasing function

(B) y(x) is a decreasing function

(C) There exists a real number β such that the line y = β intersects the curve y = y(x) at infinitely many points

(D) y(x) is a periodic function

9. If y(x) is the colution of the differential equation

$$xdy - (y^2 - 4y)\,dx = 0 \text{ for } x > 0, \quad y(1) = 2,$$

and the slope of the curve y = y(x) is never zero, then the value of $10y(\sqrt{2})$ is _____.**[JEE Advanced 2022]**

10. Let f : $[1, \infty) \to$ **R** be a differentiable function such that $f(1) = \frac{1}{3}$ and

$3\int_1^x f(t)dt = x\,f(x) - \frac{x^3}{3}$, $x \in [1, \infty)$. Let e denote the base of the natural logarithm Then the value of f(e) is:

[JEE Advanced 2023]

(A) $\frac{e^2 + 4}{3}$

(B) $\frac{\log_e 4 + e}{3}$

(C) $\frac{4e^2}{3}$

(D) $\frac{e^2 - 4}{3}$

11. For $x \in \mathbf{R}$, let y(x) be a solution of the differential equation

$(x^2 - 5)\dfrac{dy}{dx} - 2xy = -2x(x^2-5)^2$ such that $y(2) = 7$.

Then the maximum value of the function y(x) is : **[JEE Advanced 2023]**

12. Let f(x) be a continuously differentiable function on the interval $(0, \infty)$ such that $f(1) = 2$ and

$$\lim_{t \to x} \frac{t^{10}f(x) - x^{10}f(t)}{t^9 - x^9} = 1$$

for each $x > 0$. Then, for all $x > 0$, f(x) is equal to : **[JEE Advanced 2024]**

(A) $\dfrac{31}{11x} - \dfrac{9}{11}x^{10}$ (B) $\dfrac{9}{11x} + \dfrac{13}{11}x^{10}$ (C) $\dfrac{-9}{11x} + \dfrac{13}{11}x^{10}$ (D) $\dfrac{13}{11x} + \dfrac{9}{11}x^{10}$

13. For all $x > 0$, let $y_1(x)$, $y_2(x)$ and $y_3(x)$ be the functions satisfying

$$\frac{dy_1}{dx} - (\sin x)^2 y_1 = 0 \ , \ y_1(1) = 5$$

$$\frac{dy_2}{dx} - (\cos x)^2 y_2 = 0 \ , \ y_2(1) = \frac{1}{3}$$ **[JEE Advanced 2025]**

$$\frac{dy_3}{dx} - \left(\frac{2 - x^3}{x^3}\right) y_3 = 0 \ , \ y_3(1) = \frac{3}{5e}$$

respectively. Then

$$\lim_{x \to 0^+} \frac{y_1(x)y_2(x)y_3(x) + 2x}{e^{3x}\sin x}$$

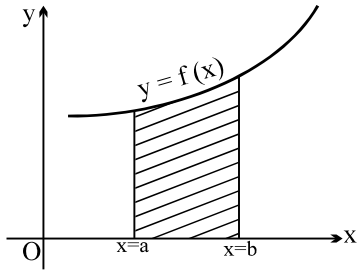
is equal to _____.

14. Let y(x) be the solution of the differential equation $x^2\dfrac{dy}{dx} + xy = x^2 + y^2$, $x > \dfrac{1}{e}$, satisfying $y(1) = 0$.

Then the value of $2\dfrac{(y(e))^2}{y(e^2)}$ is_____. **[JEE Advanced 2025]**

KEY CONCEPTS (AREA UNDER THE CURVE)

THINGS TO REMEMBER :

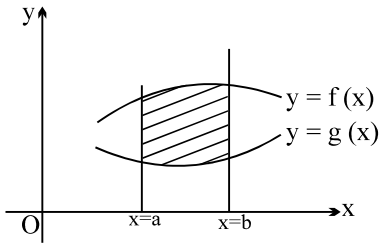


1. The area bounded by the curve y = f(x) , the x-axis and the ordinates at x = a & x = b is given by,

$$A = \int_a^b f(x)\,dx = \int_a^b y\,dx.$$

2. If the area is below the x–axis then A is negative. The convention is to consider the magnitude only i.e.

$$A = \left| \int_a^b y\,dx \right| \text{ in this case.}$$



3. Area between the curves y = f(x) & y = g(x) between the ordinates at x = a & x = b is given by,

$$A = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx = \int_a^b [f(x) - g(x)]\,dx.$$

4. Average value of a function y = f(x) w.r.t. *x* over an interval $a \le x \le b$ is defined as :

$$y(av) = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

5. The area function A_a^x satisfies the differential equation $\frac{dA_a^x}{dx} = f(x)$ with initial condition $A_a^a = 0$.

Note : If F (x) is any integral of f (x) then ,

$A_a^x = \int f(x)\,dx = F(x) + c$ $\qquad$ $A_a^a = 0 = F(a) + c \Rightarrow c = -F(a)$

hence $A_a^x = F(x) - F(a)$. Finally by taking x = b we get , $A_a^b = F(b) - F(a)$.

6. CURVE TRACING :

The following outline procedure is to be applied in Sketching the graph of a function y = f(x) which in turn will be extremely useful to quickly and correctly evaluate the area under the curves.

(a) Symmetry : The symmetry of the curve is judged as follows :

(i) If all the powers of *y* in the equation are even then the curve is symmetrical about the axis of *x*.

(ii) If all the powers of *x* are even , the curve is symmetrical about the axis of *y*.

(iii) If powers of *x* & *y* both are even, the curve is symmetrical about the axis of *x* as well as *y*.

(iv) If the equation of the curve remains unchanged on interchanging *x* and *y*, then the curve is symmetrical about y = x.

(v) If on interchanging the signs of *x* & *y* both the equation of the curve is unaltered then there is symmetry in opposite quadrants.

(b) Find dy/dx & equate it to zero to find the points on the curve where you have horizontal tangents.

(c) Find the points where the curve crosses the x–axis & also the y–axis.

(d) Examine if possible the intervals when f(x) is increasing or decreasing. Examine what happens to 'y' when $x \to \infty$ or $-\infty$.

7. USEFUL RESULTS :

(i) Whole area of the ellipse, $x^2/a^2 + y^2/b^2 = 1$ is π ab.

(ii) Area enclosed between the parabolas $y^2 = 4ax$ & $x^2 = 4by$ is $16ab/3$.

(iii) Area included between the parabola $y^2 = 4ax$ & the line $y = mx$ is $8a^2/3m^3$.

PROFICIENCY TEST-1

1. The area between the curve $xy = a^2$, x-axis, $x = a$ and $x = 2a$ is $(a > 0)$

(A) a log 2 (B) a^2 log 2 (C) 2a log 2 (D) None of these

2. Area under the curve $y = \sin 2x + \cos 2x$ between $x = 0$ and $x = \frac{\pi}{4}$, is

(A) 2 sq. units (B) 1 sq. units (C) 3 sq. units (D) 4 sq. units

3. The area bounded by the lines $y = x$, $y = 0$ and $x = 2$ is

(A) 1 (B) 2 (C) 4 (D) None of these

4. The area between the curve $y = \log x$ and x-axis which lies between $x = a$ and $x = b$ $(1 < a < b)$ is

(A) b log (b/e) – a log (a/e) (B) b log (b/e) + a log (a/e)

(C) log ab (D) log (b/a)

5. Area bounded by the curve $y = xe^{x^2}$, x- axis and the ordinates $x = 0$, $x = \alpha$ is $(\alpha > 0)$

(A) $\frac{e^{\alpha^2}+1}{2}$ sq. units (B) $\frac{e^{\alpha^2}-1}{2}$ sq.units (C) $e^{\alpha^2}+1$ sq. units (D) $e^{\alpha^2}-1$ sq.units

6. The area bounded between the curve $y = 2x^2 + 5$, x-axis and ordinates $x = -2$ and $x = 1$ is

(A) 21 (B) 29/5 (C) 23 (D) 24

7. The area bounded by the curve $y = x \sin x^2$, x-axis, $x = 0$ and $x = \sqrt{\frac{\pi}{2}}$ is

(A) 1/2 (B) $1/\sqrt{2}$ (C) 1/4 (D) $\pi/2$

8. The area bounded between the curve $\frac{x}{4} - \frac{y}{2} + 1 = 0$, $x = -2$, $x = 3$ and x-axis is

(A) 45/4 (B) 45/2 (C) 15 (D) 25/2

9. The area bounded by the curve $y = \frac{2}{1+\cos 2x}$, coordinate axes and $x = \pi/4$ is

(A) 1 (B) 2 (C) $\pi/4$ (D) Not defined

10. The area enclosed between the curve $x = 2y - y^2$ and y-axis is

(A) 9/4 (B) 4/3 (C) 9 (D) None of these

11. The area between the curve $y = \sin^3 x$, x-axis, and $x = \pi/2$ in first quadrant is

(A) 1 (B) 1/3 (C) 2/3 (D) 3/2

12. The value of a for which the area of the region bounded by the curve $y = \sin 2x$, the straight lines $x = \pi/6$, $x = a$ and x-axis is equal to 1/2 is $(a > \pi/6)$

(A) $\pi/2$ (B) $\pi/3$ (C) $2\pi/3$ (D) None of these

PROFICIENCY TEST-2

1. The area bounded by the curve $y = 4x^2$, $y = 1$ and $y = 4$ in the first quadrant is

(A) $2\frac{2}{3}$ (B) $3\frac{1}{3}$ (C) $2\frac{1}{3}$ (D) $3\frac{1}{2}$

2. The area between the curve $y = \sec x$ and y-axis when $1 \le y \le 2$ is

(A) $\frac{2\pi}{3} - \log (2 + \sqrt{3})$ (B) $\frac{2\pi}{3} + \log (2 + \sqrt{3})$

(C) $\frac{\pi}{3} - \frac{1}{2}\log (2 + \sqrt{3})$ (D) None of these

3. Area of the enclosed region bounded by the curve $y = x (x - 1)^2$ and x-axis is
(A) 4 (B) 1/3 (C) 1/12 (D) 1/2

4. The area bounded by the curve $y = \log(x)$, coordinate axes and $y = 1$ is
(A) e (B) e + 1 (C) e – 1 (D) 1/e

5. The area bounded between the curve $|y| = 1 - x^2$ is
(A) 2/3 (B) 4/3 (C) 8/3 (D) None of these

6. The whole area bounded by the curves $x = a\cos t$, $y = b\sin t$ is (a, b > 0)

(A) πab (B) $\frac{\pi}{2}ab$ (C) $\frac{\pi}{4}ab$ (D) None of these

7. The area between the curve $y = \tan x$ and x-axis, when $-\pi/4 \le x \le \pi/4$ is
(A) log 2 (B) log 4 (C) log $\sqrt{2}$ (D) None of these

8. The area bounded between parabola $x^2 = 4y$ and $y = |x|$ is
(A) 2/3 (B) 4/3 (C) 8/3 (D) 16/3

9. The common area of the curves $y = \sqrt{x}$ and $x = \sqrt{y}$ is
(A) 3 (B) 5/3 (C) 1/3 (D) None of these

10. Area of the region bounded by the curves $y = e^x$, $y = e^{-x}$ and the straight line $y = 2$ is
(A) log (4/e) (B) 2 log (4/e) (C) 4 log (4/e) (D) None of these

11. The area bounded by $y = \tan x$, $y = \cot x$, x-axis in $0 \le x \le \frac{\pi}{2}$ is
(A) log 2 (B) 3 log 2 (C) 2 log 2 (D) 4 log 2

12. The area bounded by the curve $y = 2x - x^2$ and straight line $y = -x$ is
(A) 35/6 (B) 9/2 (C) 43/6 (D) None of these

13. Common area between the parabolas $y = 2x^2$ and $y = x^2 + 4$ is
(A) 16/3 (B) 8/3 (C) 32/3 (D) None of these

14. If A is the area between the curve $y = \sin x$ and x-axis in the interval $[0, \pi/2]$, then the area between $y = \sin 2x$ and x-axis in this interval will be
(A) A (B) 2A (C) A/2 (D) None of these

15. Find the area enclosed by the lines $2y = x$, $y = 2x$ and $x = 4$ is
(A) 1 (B) 2 (C) 12 (D) 16

EXERCISE–I

1. Compute the area of the region bounded by the curves y = e. x. *ln x* & y = *ln x*/(e. *x)* where *ln* e=1.

2. A figure is bounded by the curves y = $\left|\sqrt{2}\sin\frac{\pi x}{4}\right|$, y = 0, x = 2 & x = 4. At what angles to the positive x-axis straight lines must be drawn through (4 , 0) so that these lines partition the figure into three parts of the same size.

3. If the area enclosed by the parabolas $y = a - x^2$ and $y = x^2$ is $18\sqrt{2}$ sq. units. Find the value of 'a'.

4. The line 3x + 2y = 13 divides the area enclosed by the curve,
$9x^2 + 4y^2 - 18x - 16y - 11 = 0$ into two parts. Find the ratio of the larger area to the smaller area.

5. Find the area of the region enclosed between the two circles $x^2 + y^2 = 1$ & $(x-1)^2 + y^2 = 1$

6. Find the values of m (m > 0) for which the area bounded by the line y = mx + 2 and $x = 2y - y^2$ is , (i) 9/2 square units & (ii) minimum. Also find the minimum area.

7. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let D_1 denotes the region surrounded by C_1, C_2 and the line x = 1 and D_2 denotes the region surrounded by C_1, C_2 and the line x = a. If $D_1 = D_2$. Find the value of 'a'.

8. Find the value (s) of the parameter 'a' (a > 0) for each of which the area of the figure bounded by the straight line, $y = \frac{a^2 - ax}{1 + a^4}$ & the parabola $y = \frac{x^2 + 2ax + 3a^2}{1 + a^4}$ is the greatest.

9. Find the value of 'c' for which the area of the figure bounded by the curve, $y = 8x^2 - x^5$, the straight lines x = 1 & x = c & the abscissa axis is equal to 16/3.

10. Compute the area included between the straight lines , $x - 3y + 5 = 0$; $x + 2y + 5 = 0$ and the circle $x^2 + y^2 = 25$.

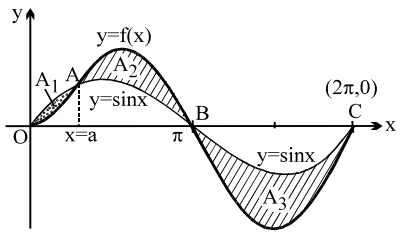
11. A polynomial function *f* (x) satisfies the condition *f* (x + 1) = *f* (x) + 2x + 1. Find *f* (x) if *f* (0) = 1. Find also the equations of the pair of tangents from the origin on the curve y = *f* (x) and compute the area enclosed by the curve and the pair of tangents.

12. Find the equation of the line passing through the origin and dividing the curvilinear triangle with vertex at the origin , bounded by the curves $y = 2x - x^2$, y = 0 and *x* = 1 into two parts of equal area.

13. Consider the curve $y = x^n$ where n > 1 in the 1st quadrant. If the area bounded by the curve, the x-axis and the tangent line to the graph of $y = x^n$ at the point (1, 1) is maximum then find the value of n.

14. Consider the collection of all curve of the form $y = a - bx^2$ that pass through the the point (2, 1), where *a* and *b* are positive constants. Determine the value of *a* and *b* that will minimise the area of the region bounded by $y = a - bx^2$ and x-axis. Also find the minimum area.

15. In the adjacent figure, graphs of two functions y = f(x) and y = sinx are given. y = sinx intersects, y = f(x) at A (a, f(a)); B(π, 0) and C(2π, 0). A_i (i = 1, 2, 3,) is the area bounded by the curves y = f (x) and y = sinx between x=0 and x= a; i = 1, between x = a and x = π; i = 2, between x = π and x = 2π; i = 3. If $A_1 = 1 - \sin a + (a-1)\cos a$, determine the function f(x). Hence determine 'a' and A_1. Also calculate A_2 and A_3.



16. Show that the area bounded by the curve $y = \frac{\ln x - c}{x}$, the x-axis and the vertical line through the maximum point of the curve is independent of the constant c.

17. For what value of 'a' is the area of the figure bounded by the lines,
$y = \frac{1}{x}$, $y = \frac{1}{2x-1}$, x = 2 & x = a equal to $\ln \frac{4}{\sqrt{5}}$?

18. Compute the area of the loop of the curve $y^2 = x^2 [(1 + x)/(1 - x)]$.

19. For the curve $f(x) = \frac{1}{1+x^2}$, let two points on it are $A(\alpha, f(\alpha))$, $B\left(-\frac{1}{\alpha}, f\left(-\frac{1}{\alpha}\right)\right)$ ($\alpha > 0$). Find the minimum area bounded by the line segments OA, OB and $f(x)$, where 'O' is the origin.

20. Let 'c' be the constant number such that c > 1. If the least area of the figure given by the line passing through the point (1, c) with gradient 'm' and the parabola $y = x^2$ is 36 sq. units find the value of $(c^2 + m^2)$.

21. Let A_n be the area bounded by the curve $y = (\tan x)^n$ & the lines $x = 0$, $y = 0$ & $x = \pi/4$.
Prove that for $n > 2$, $A_n + A_{n-2} = 1/(n-1)$ & deduce that $1/(2n+2) < A_n < 1/(2n-2)$.

22. If f (x) is monotonic in (a, b) then prove that the area bounded by the ordinates at x = a ; x = b ; y = f (x) and y = f (c), $c \in (a, b)$ is minimum when $c = \frac{a+b}{2}$.

Hence if the area bounded by the graph of $f(x) = \frac{x^3}{3} - x^2 + a$, the straight lines x = 0, x = 2 and the x-axis is minimum then find the value of 'a'.

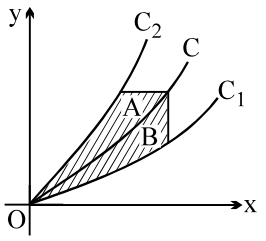
23. Consider the two curves $C_1 : y = 1 + \cos x$ & $C_2 : y = 1 + \cos(x - \alpha)$ for $\alpha \in (0, \pi/2)$; $x \in [0, \pi]$. Find the value of α, for which the area of the figure bounded by the curves C_1, C_2 & x = 0 is same as that of the figure bounded by C_2, y = 1 & $x = \pi$. For this value of α, find the ratio in which the line y = 1 divides the area of the figure by the curves C_1, C_2 & $x = \pi$.

24. Compute the area of the figure which lies in the first quadrant inside the curve $x^2 + y^2 = 3a^2$ & is bounded by the parabola $x^2 = 2ay$ & $y^2 = 2ax$ ($a > 0$).

25. Find the whole area included between the curve $x^2y^2 = a^2(y^2 - x^2)$ & its asymptotes (asymptotes are the lines which meet the curve at infinity).

26. For what values of $a \in [0, 1]$ does the area of the figure bounded by the graph of the function y = f (x) and the straight lines x = 0, x = 1 & y = f(a) is at a minimum & for what values it is at a maximum if $f(x) = \sqrt{1-x^2}$. Find also the maximum & the minimum areas.

27. Let C_1 & C_2 be two curves passing through the origin as shown in the figure. A curve C is said to "bisect the area" the region between C_1 & C_2, if for each point P of C, the two shaded regions A & B shown in the figure have equal areas. Determine the upper curve C_2, given that the bisecting curve C has the equation $y = x^2$ & that the lower curve C_1 has the equation $y = x^2/2$.



28. Given $f(x) = \int_0^x e^t (\ln \sec t - \sec^2 t)\, dt$; $g(x) = -2e^x \tan x$. Find the area bounded by the curves y = f (x) and y = g (x) between the ordinates x = 0 and $x = \frac{\pi}{3}$.

EXERCISE–II

1. The area between the curve $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the straight line $\frac{x}{a}+\frac{y}{b}=1$, in the first quadrant is $(a, b > 0)$

(A) $\frac{1}{4}\pi ab - \frac{1}{2}ab$ (B) $\frac{1}{4}ab$ (C) $\frac{1}{2}\pi ab$ (D) None of these

2. The area of the region consisting of all points (x, y) so that $x^2 + y^2 \le 1 \le |x| + |y|$, is (in sq. units)

(A) π (B) $\pi - 1$ (C) $\pi - 2$ (D) $\pi - 3$

3. The area of the region defined by the inequality $|x| + |y| + |x + y| \le 1$ is (in sq. units)

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{4}$

4. If the area enclosed between $f(x) = \min\left(\cos^{-1}(\cos x), \cot^{-1}(\cot x)\right)$ and x-axis in $x \in (\pi, 2\pi)$ is $\frac{\pi^2}{k}$ sq. units, where $k \in N$, then k is equal to

(A) 4 (B) 6 (C) 8 (D) 12

5. If the area enclosed by $y^2 = 4ax$ and line $y = ax$ is $\frac{1}{3}$ sq. unit, then the area enclosed by $y = 4x$ with same parabola is (in sq. units)

(A) 8 (B) 4 (C) $\frac{4}{3}$ (D) $\frac{8}{3}$

6. The smaller area enclosed by $y = f(x)$, when $f(x)$ is a polynomial of least degree satisfying $\lim_{x\to 0}\left(1+\frac{f(x)}{x^3}\right)^{1/x} = e$, and the circle $x^2 + y^2 = 2$, is (in sq. units)

(A) $\frac{\pi}{2}$ (B) $\frac{3}{5}$ (C) $\frac{\pi}{2}-\frac{3}{5}$ (D) $\frac{\pi}{2}+\frac{3}{5}$

7. The area bounded by the curve $f(x) = x + \sin x$ and its inverse function betweeen the ordinates $x = 0$ and $x = 2\pi$ is (in sq. units)

(A) 4π (B) 8π (C) 4 (D) 8

8. If $f(x) = a + bx + cx^2$, where $c > 0$ and $b^2 - 4ac < 0$, then the area enclosed by the co-ordinate axes, the line $x = 2$ and the curve $y = f(x)$ is given by (in sq. units)

(A) $\frac{1}{3}[4f(1) + f(2)]$ (B) $\frac{1}{2}[f(0) + 4f(1)]$

(C) $\frac{1}{2}[f(0) + 4f(1) + f(2)]$ (D) $\frac{1}{3}[f(0) + 4f(1) + f(2)]$

9. Area of region bounded by $[x]^2 = [y]^2$ if $x \in [1, 5]$ (where $[\cdot]$ represents the greatest integer function), is (in sq. units)

(A) 10 (B) 8 (C) 6 (D) 4

10. The area bounded by the curve $y = x^2 + 2x + 1$, tangent to the curve at (1, 4) and y-axis is (in sq. units)

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) 2 (D) 1

11. Let $f(x) = \begin{cases} x^2, & x < 0 \\ x, & x \geq 0 \end{cases}$; area bounded by the curve y = f(x), y = 0 and x = ±3a is $\frac{9a}{2}$ sq unit, a > 0.

Then a is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) 1

12. Area bounded by the curve $xy^2 = a^2(a - x)$ and y-axis (in sq. units), given $a \neq 0$, is :

(A) $\frac{\pi a^2}{2}$ (B) πa^2 (C) $2\pi a^2$ (D) Not defined

13. For $x \in \left[0, \frac{\pi}{3}\right]$, α equals to area between the curves y = cosx and y = cos2x ; β equals to area between the curve y = cos2x and x-axis ; then $(\alpha + \beta)$ equals to (in sq. units)

(A) 1 (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) 2

14. The area of the region satisfying $|y| + \frac{1}{2} \leq e^{-|x|}$, (in sq. units), is

(A) (1 – log2) (B) 2(1 – log2) (C) e(1 – log2) (D) 4(1 – log2)

15. Let $f(x) = \min\left\{e^x, \frac{3}{2}, 1 + e^{-x}\right\}$, $\forall\, x \in R$. The area bounded by the curve y = f(x), co-ordinate axes and the line x = 1, is $\left(2 + \ln\frac{a}{b\sqrt{b}} - \frac{1}{e}\right)$ sq. units, where, a & b are single-digit natural numbers, then, (a + b) equals :

(A) 4 (B) 5 (C) 7 (D) 8

EXERCISE–III

1. If the area bounded by the x- axis, curve y = f(x) and the lines x = 1, x = b is equal to $\sqrt{b^2+1} - \sqrt{2}$ for all b > 1, then f(x) is **[AIEEE 2002]**

(A) $\sqrt{(x-1)}$ (B) $\sqrt{(x+1)}$ (C) $\sqrt{(x^2+1)}$ (D) $\frac{x}{\sqrt{1+x^2}}$

2. The area of the region bounded by the curves y = | x – 1| and y = 3 – |x| is- **[AIEEE 2003]**

(A) 6 sq. units (B) 2 sq. units (C) 3 sq. units (D) 4 sq. units

3. The area of the region bounded by the curves y = |x – 2|, x = 1, x = 3 and the x- axis is-

(A) 1 (B) 2 (C) 3 (D) 4 **[AIEEE 2004]**

4. The area enclosed between the curve $y = \log_e(x + e)$ and the coordinate axes is - **[AIEEE-2005]**

(A) 1 (B) 2 (C) 3 (D) 4

5. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines x = 4, y = 4 and the coordinate axes. If S_1, S_2, S_3 are respectively the areas of these parts numbered from top to bottom; then $S_1 : S_2 : S_3$ is - **[AIEEE-2005]**

(A) 1 : 2 : 1 (B) 1 : 2 : 3 (C) 2 : 1 : 2 (D) 1 : 1 : 1

6. Let f (x) be a non-negative continuous function such that the area bounded by the curve y = f (x), x-axis and the ordinates x = $\frac{\pi}{4}$ and x = $\beta > \frac{\pi}{4}$ is $\left(\beta \sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta\right)$. Then f $\left(\frac{\pi}{2}\right)$ is - **[AIEEE-2005]**

(A) $\left(\frac{\pi}{4} + \sqrt{2} - 1\right)$ (B) $\left(\frac{\pi}{4} - \sqrt{2} + 1\right)$ (C) $\left(1 - \frac{\pi}{4} - \sqrt{2}\right)$ (D) $\left(1 - \frac{\pi}{4} + \sqrt{2}\right)$

7. The area enclosed between the curves $y^2 = x$ and $y = |x|$ is **[AIEEE 2007]**

(A) 2/3 (B) 1 (C) 1/6 (D) 1/3

8. The area of the plane region bounded by the curves $x + 2y^2 = 0$ and $x + 3y^2 = 1$ is equal to -**[AIEEE 2008]**

(A) 1/3 (B) 2/3 (C) 4/3 (D) 5/3

9. The area of the region bounded by the parabola $(y - 2)^2 = x - 1$, the tangent to the parabola at the point (2, 3) and the x – axis is - **[AIEEE 2009]**

(A) 3 (B) 6 (C) 9 (D) 12

10. The area bounded by the curves y = cos x and y = sin x between the ordinates x = 0 and x = $\frac{3\pi}{2}$ is

(A) $4\sqrt{2}+1$ (B) $4\sqrt{2}-2$ (C) $4\sqrt{2}+2$ (D) $4\sqrt{2}-1$ **[AIEEE-2010]**

11. The area of the region enclosed by the curves y = x, x = e, y = $\frac{1}{x}$ and the positive x-axis is **[AIEEE-2011]**

(A) $\frac{1}{2}$ square units (B) 1 square units (C) $\frac{3}{2}$ sqaure units (D) $\frac{5}{2}$ square units

12. The area bounded between the parabolas $x^2 = \frac{y}{4}$ and $x^2 = 9y$ and the straight line y = 2 is

(A) $20\sqrt{2}$ (B) $\frac{10\sqrt{2}}{3}$ (C) $\frac{20\sqrt{2}}{3}$ (D) $10\sqrt{2}$ **[AIEEE-2012]**

13. The area (in square units) bounded by the curves y = $\sqrt{x}$, 2y – x + 3 = 0, x-axis and lying in the first quadrant is : **[JEE Main 2013]**

(A) 27/4 (B) 9 (C) 36 (D) 18

14. The area of the region described by A = {(x, y) : $x^2 + y^2 \le 1$ and $y^2 \le 1 - x$} is **[JEE Main 2014]**

(A) $\frac{\pi}{2} + \frac{2}{3}$ (B) $\frac{\pi}{2} + \frac{4}{3}$ (C) $\frac{\pi}{2} - \frac{4}{3}$ (D) $\frac{\pi}{2} - \frac{2}{3}$

15. The area (in sq. units) of the region described by **[JEE Main 2015]**
{(x,y): $y^2 \le 2x$ and $y \ge 4x - 1$} is :

(A) $\frac{9}{32}$ (B) $\frac{7}{32}$ (C) $\frac{5}{64}$ (D) $\frac{15}{64}$

16. The area (in sq. units) of the region {(x, y) : $y^2 \ge 2x$ and $x^2 + y^2 \le 4x$, $x \ge 0$, $y \ge 0$} is : **[JEE Main 2016]**

(A) $\pi - \frac{4}{3}$ (B) $\pi - \frac{8}{3}$ (C) $\pi - \frac{4\sqrt{2}}{3}$ (D) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$

17. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is **[JEE Main 2017]**

(A) $\frac{7}{3}$ (B) $\frac{5}{2}$ (C) $\frac{59}{12}$ (D) $\frac{3}{2}$

18. Let $g(x) = \cos x^2$, $f(x) = \sqrt{x}$, and α, β $(\alpha < \beta)$ be the roots of the quadratic equation $18x^2 - 9\pi x + \pi^2 = 0$. Then the area (in sq. uinits) bounded by the curve $y = (gof)(x)$ and the lines $x = \alpha$, $x = \beta$ and $y = 0$, is : **[JEE Main 2018]**

(A) $\frac{1}{2}\left(\sqrt{2}-1\right)$ (B) $\frac{1}{2}\left(\sqrt{3}-1\right)$ (C) $\frac{1}{2}\left(\sqrt{3}+1\right)$ (D) $\frac{1}{2}\left(\sqrt{3}-\sqrt{2}\right)$

EXERCISE–IV

1. Area of the region $\left\{(x, y) \in \mathbb{R}^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ is equal to **[IIT JEE (Adv.) 2016]**

(A) $\frac{1}{6}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{3}$

2. If the line $x = \alpha$ divides the area of region $R = \{(x, y) \in \mathbb{R}^2 : x^3 \leq y \leq x, 0 \leq x \leq 1\}$ into two equal parts, then **[IIT JEE Advance - 2017]**

(A) $\frac{1}{2} < \alpha < 1$ (B) $\alpha^4 + 4\alpha^2 - 1 = 0$ (C) $0 < \alpha \leq \frac{1}{2}$ (D) $2\alpha^4 - 4\alpha^2 + 1 = 0$

3. A farmer F_1 has a land in the shape of a triangle with vertices at P(0, 0), Q(1, 1) and R(2, 0). From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ $(n > 1)$. If the area of the region taken away by the farmer F_2 is exactly 30% of the area of ΔPQR, then the value of n is _____ . **[JEE Advanced 2018]**

4. Let $f : [0, \infty) \to R$ be a continuous function such that **[JEE Advanced 2018]**

$$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$$

for all $x \in [0, \infty)$. Then, which of the following statement(s) is(are) TRUE?

(A) The curve $y = f(x)$ passes through the point (1, 2)

(B) The curve $y = f(x)$ passes through the point (2, –1)

(C) The area of the region $\{(x, y) \in [0, 1] \times R : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-2}{4}$

(D) The area of the region $\{(x, y) \in [0, 1] \times R : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-1}{4}$

5. The area of the region $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ is **[JEE Advanced 2019]**

(A) $16\log_e 2 - 6$ (B) $8\log_e 2 - \frac{7}{3}$ (C) $16\log_e 2 - \frac{14}{3}$ (D) $8\log_e 2 - \frac{14}{3}$

6. Let the functions $f : R \to R$ and $g : R \to R$ be defined by **[JEE Advanced 2020]**

$f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}\left(e^{x-1} + e^{1-x}\right)$.

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is

(A) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(B) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e-e^{-1}\right)$

(C) $\left(2-\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

(D) $\left(2+\sqrt{3}\right)+\frac{1}{2}\left(e+e^{-1}\right)$

7. Consider all rectangles lying in the region $\left\{(x,y) \in R \times R: 0 \le x \le \frac{\pi}{2} \text{ and } 0 \le y \le 2\sin(2x)\right\}$ and having one side on the x-axis. The area of the rectangle which has the maximum perimeter among all such rectangles, is : **[JEE Advanced 2020]**

(A) $\frac{3\pi}{2}$ (B) π (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi\sqrt{3}}{2}$

8. The area of the region $\left\{(x, y) : 0 \le x \le \frac{9}{4}, 0 \le y \le 1, x \ge 3y, x + y \ge 2\right\}$ is **[JEE Advanced 2021]**

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

9. Consider the function $f, g : \mathbf{R} \to \mathbf{R}$ defined by

$$f(x) = x^2 + \frac{5}{12} \quad \text{and} \quad g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right), & |x| \le \frac{3}{4} \\ 0, & |x| > \frac{3}{4} \end{cases}$$

If α is the area of the region

$$\left\{(x,y) \in R \times R : |x| \le \frac{3}{4},\ 0 \le y \le \min\{f(x), g(x)\}\right\},$$

then the value of 9α is _____. **[JEE Advanced 2022]**

10. Let f : [0, 1] $\to$ [0,1] be the function defined by $f(x) = \frac{x^3}{3} - x^2 + \frac{5}{9}x + \frac{17}{36}$. Consider the square region S = [0, 1] × [0, 1]. Let G = {(x, y) $\in$ S : y > f(x)} be called the green region and R = {(x, y) $\in$ S : y < f(x)} be called the red region. Let L_h = {(x, h) $\in$ S : x $\in$ [0, 1]} be the horizontal line drawn at a height h $\in$ [0, 1]. Then which of the following statements is(are) true? **[JEE Advanced 2023]**

(A) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h

(B) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h

(C) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h

(D) There exists an $h \in \left[\frac{1}{4}, \frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h

11. Let n $\geq$ 2 be a natural number and f : [0, 1] $\to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} n(1-2nx) & \text{if } 0 \leq x \leq \frac{1}{2n} \\ 2n(2nx-1) & \text{if } \frac{1}{2n} \leq x \leq \frac{3}{4n} \\ 4n(1-nx) & \text{if } \frac{3}{4n} \leq x \leq \frac{1}{n} \\ \frac{n}{n-1}(nx-1) & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$

If n is such that the area of the region bounded by the curves x = 0 , x = 1 , y = 0 and y = f(x) is 4 , then the maximum value of the function f is : **[JEE Advanced 2023]**

12. Let S = {(x, y) $\in \mathbb{R} \times \mathbb{R}$: x $\geq$ 0, y $\geq$ 0, $y^2 \leq 4x$, $y^2 \leq 12 - 2x$ and $3y + \sqrt{8}x \leq 5\sqrt{8}$ }. If the area of the region S is $\alpha\sqrt{2}$, then α is equal to : **[JEE Advanced 2024]**

(A) $\frac{17}{2}$ (B) $\frac{17}{3}$ (C) $\frac{17}{4}$ (D) $\frac{17}{5}$

13. Let $\mathbb{R}$ denote the set of all real numbers. Then the area of the region $\left\{(x,y) \in \mathbb{R} \times \mathbb{R} : x > 0, y > \frac{1}{x}, 5x - 4y - 1 > 0, 4x + 4y - 17 < 0\right\}$ is **[JEE Advanced 2025]**

(A) $\frac{17}{16} - \log_e 4$ (B) $\frac{33}{8} - \log_e 4$ (C) $\frac{57}{8} - \log_e 4$ (D) $\frac{17}{2} - \log_e 4$

ANSWER KEY

DIFFERENTIAL EQUATION

EXERCISE–I

1. **(i)** order 2 & degree 3 **(ii)** order 2 & degree 2

2. $[1 + (y')^2] .y''' - 3y'(y'')^2 = 0$

3. $12y (y')^2 = x[8(y')^3 - 27]$

4. $\ell n^2 (\sec x + \tan x) - \ell n^2 (\sec y + \tan y) = c$

5. $\sqrt{x^2 - 1} - \sec^{-1} x + \sqrt{y^2 - 1} = c$

6. $y = c (1 - ay) (x + a)$

7. $\ln\left[1 + \tan\frac{x + y}{2}\right] = x + c$

8. $y \sin y = x^2 \ell n x + c$

9. $m = m_0 e^{-kt}$ where $k = -\frac{1}{t_0}\ln\left(1 - \frac{\alpha}{100}\right)$

10. (a) $P = 1000 + 1500e^{-kt}$ where $k = \frac{1}{10}\ell n\left(\frac{5}{3}\right)$; (b) $T = 10 \log_{5/3}(3)$; (c) $P = 1000$ as $t \to \infty$

11. $y = (x + 1) . \ln (x + 1) - x + 3$

12. $x^2 + y^2 = k^2$

13. $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

14. $y = x^{1/n}$

15. $\sqrt{x^2 - y^2} + \sqrt{1 + x^2 - y^2} = \frac{c (x + y)}{\sqrt{x^2 - y^2}}$

16. $y = kx$ or $xy = c$

EXERCISE–II

1. **(a)** $c(x - y)^{2/3} (x^2 + xy + y^2)^{1/6} = \exp\left[\frac{1}{\sqrt{3}}\tan^{-1}\frac{x + 2y}{x\sqrt{3}}\right]$ where $\exp x \equiv e^x$ **(b)** $y^2 - x^2 = c (y^2 + x^2)^2$

2. $\frac{y^2 \pm y\sqrt{y^2 - x^2}}{x^2} = \ell n\left|\left(y \pm \sqrt{y^2 - x^2}\right)\cdot\frac{c^2}{x^3}\right|$, where same sign has to be taken.

4. $x^2 + y^2 - 2x = 0$

6. $\frac{1}{2}\ell n\left|x^2 + a^2\right| - \tan^{-1}\left(\frac{a}{x}\right) = c$, where $a = x + y^2$

7. $xy \cos\frac{y}{x} = c$

8. $x^2 + y^2 = cx$

9. $\text{arc tan}\frac{2y + 1}{2x + 1} = \ell n\, c\sqrt{x^2 + y^2 + x + y + \frac{1}{2}}$

10. $(x + y - 2) = c (y - x)^3$

11. $\tan^{-1}\frac{y + 3}{x + 2} + \ell n\, c\sqrt{(y + 3)^2 + (x + 2)^2} = 0$

12. $x + y + \frac{4}{3} = ce^{3(x-2y)}$

13. $e^{-2\tan^{-1}\frac{y + 2}{x - 3}} = c . (y + 2)$

14. $(\cos y - \sin x - 1)^2 (\cos y + \sin x - 1)^5 = c$

EXERCISE–III

1. $y = 2(e^x - x - 1)$

2. $y = cx^2 \pm x$

3. $y(x-1) = x^2(x^2 - x + c)$

4. $xy = c - \arctan x$

5. $y = cx - x^2$

6. $\left(\frac{1}{3} + y\right)\tan^3\frac{x}{2} = c + 2\tan\frac{x}{2} - x$

7. $y = cx + x\,\ell n \tan x$

8. $x = ce^{-\arctan y} + \arctan y - 1$

9. $y = cx \pm \frac{a^2}{2x}$

11. $\cos x - 1$

12. $f(x) = -\frac{2\cos x}{(1+\sin x)^2} - Ce^{-\sin x}\cdot\cos x$

14. $x = \ell ny\left(cx^2 + \frac{1}{2}\right)$

15. $e^{-x^2/2} = y(c + \cos x)$

16. $\frac{1}{y^2} = -1 + (c + x)\cot\left(\frac{x}{2} + \frac{\pi}{4}\right)$

17. $x^3 y^{-3} = 3\sin x + c$

18. $y^{-1}e^x = c - x^2$

19. $x = cy \pm \frac{a^2}{y}$

20. $27\frac{7}{9}$ minutes

21. $y^2 = 3x^2 - 6x - x^3 + ce^{-x} + 4$

22. $\sin y = (e^x + c)(1 + x)$

23. $y = ce^x$; $y = c + \frac{x^2}{2}$

24. $y = \frac{1}{x}\tan\left(\ell n\,|cx|\right)$

25. $y^2 = \frac{2}{3}\sin x + \frac{c}{\sin^2 x}$

EXERCISE–IV

1.	C	**2.**	B	**3.**	C	**4.**	C	**5.**	C	**6.**	B	**7.**	C	**8.**	C	**9.**	C	**10.**	C
11.	B	**12.**	C	**13.**	D	**14.**	B	**15.**	C	**16.**	B	**17.**	A	**18.**	D	**19.**	B	**20.**	D
21.	D	**22.**	B	**23.**	D	**24.**	A	**25.**	A	**26.**	D	**27.**	C	**28.**	B	**29.**	C	**30.**	D
31.	D	**32.**	A	**33.**	D	**34.**	D	**35.**	Bonus			**36.**	C	**37.**	A	**38.**	3.00		

EXERCISE–V

1.	B, C	**2.**	A, D	**3.**	B	**4.**	0.4	**5.**	BC	**6.**	AC	**7.**	AC
8.	C	**9.**	8.00	**10.**	C	**11.**	16.00	**12.**	B	**13.**	2.00	**14.**	0.75

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AREA UNDER THE CURVE

PROFICIENCY TEST-1

1.	B	**2.**	B	**3.**	B	**4.**	A	**5.**	B	**6.**	A	**7.**	A
8.	A	**9.**	A	**10.**	B	**11.**	C	**12.**	B				

PROFICIENCY TEST-2

1.	C	**2.**	A	**3.**	C	**4.**	C	**5.**	C	**6.**	A	**7.**	A
8.	D	**9.**	C	**10.**	B	**11.**	A	**12.**	B	**13.**	C	**14.**	A
15.	C												

EXERCISE–I

1. $(e^2 - 5)/4$ e sq. units **2.** $\pi - \tan^{-1}\frac{2\sqrt{2}}{3\pi}$; $\pi - \tan^{-1}\frac{4\sqrt{2}}{3\pi}$ **3.** a = 9

4. $\frac{3\pi + 2}{\pi - 2}$ **5.** $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$ sq. units **6.** **(i)** m = 1, **(ii)** m = ∞ ; A_{min}= 4/3

7. e **8.** $a = 3^{1/4}$ **9.** C = – 1 or $\left(8 - \sqrt{17}\right)^{1/3}$

10. $\frac{5}{4}(5\pi + 14)$ sq. units **11.** $f(x) = x^2 + 1$; $y = \pm 2x$; A = $\frac{2}{3}$ sq. units **12.** $y = 2x/3$

13. $\sqrt{2}+1$ **14.** b = 1/8, $A_{minimum} = 4\sqrt{3}$ sq. units

15. f(x) = x sinx, a = 1; $A_1 = 1 - \sin 1$; $A_2 = \pi - 1 - \sin 1$; $A_3 = (3\pi - 2)$ sq. units

16. 1/2 **17.** a = 8 or $\frac{2}{5}\left(6 - \sqrt{21}\right)$ **18.** $2 - (\pi/2)$ sq. units

19. $\frac{(\pi - 1)}{2}$ **20.** 104

22. $a = \frac{2}{3}$ **23.** $\alpha = \pi/3$, ratio = $2 : \sqrt{3}$

24. $\left[\frac{\sqrt{2}}{3} + \frac{3}{2}.\arcsin\frac{1}{3}\right]a^2$ sq. units **25.** $4a^2$

26. a = 1/2 gives minima, $A\left(\frac{1}{2}\right) = \frac{3\sqrt{3} - \pi}{12}$; a = 0 gives local maxima A(0) = $1 - \frac{\pi}{4}$;

a = 1 gives maximum value , $A(1) = \pi/4$

27. $(16/9)x^2$ **28.** $e^{\pi/3}\log 2$ sq. units

EXERCISE–II

1.	A	**2.**	C	**3.**	B	**4.**	A	**5.**	D	**6.**	D	**7.**	D
8.	D	**9.**	B	**10.**	B	**11.**	A	**12.**	B	**13.**	A	**14.**	B
15.	C												

EXERCISE–III

1.	D	**2.**	D	**3.**	A	**4.**	A	**5.**	D	**6.**	D	**7.**	C
8.	C	**9.**	C	**10.**	B	**11.**	C	**12.**	C	**13.**	B	**14.**	B
15.	A	**16.**	B	**17.**	B	**18.**	B						

EXERCISE–IV

1.	C	**2.**	A,D	**3.**	4	**4.**	BC	**5.**	C	**6.**	A	**7.**	C
8.	A	**9.**	6.00	**10.**	B,C,D	**11.**	8.00	**12.**	B	**13.**	B		